Data sets of motor imagery EEG

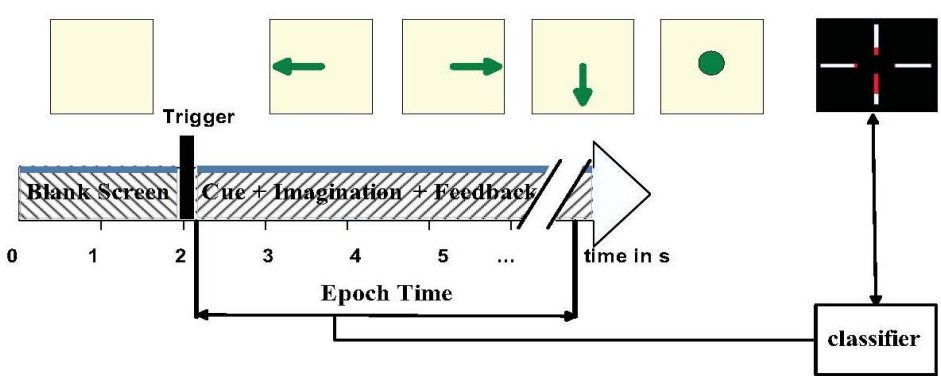
Data sets provided by the Dr. Cichocki's Lab (Lab. for Advanced Brain Signal Processing), BSI, RIKEN collaboration with Shanghai Jiao Tong University.

Experimental paradigm

This data sets of EEG data were recorded from several healthy subjects. The cue-based BCI paradigm consisted of two/three motor imagery tasks, namely the imagination of movement of the left hand (LH), right hand (RH) and both feet (F). Several sessions on different days were recorded for some subjects, the data of each session was stored in one data file respectively.

The subjects were sitting in a comfortable armchair in front of a computer screen. At the beginning of a trial, the screen is blank. After two seconds (t=2s), a cue in the form of an arrow pointing either to the left, right or down (corresponding to three classes of LH, RH and F) appeared and stayed on the screen for a specific duration (3-10 sec). This prompted the subjects to perform the desired motor imagery task. The subjects were requested to carry out the motor imagery task until the cue disappeared from the screen and try to avoid the eye blinking or eye movements during the imagination. A 2 seconds break followed when the cue is disappeared. This procedure is repeated 30-100 times for each run with the random cue sequence. The paradigm is illustrated in Figure 1. For each subject, the first run is called initialization procedure which only presents the cues without any feedback. Based on the online BCI classifier trained on the EEG data recorded from the initialization run, the system is able to present the system feedback online by several red bars representing the classification output for left hand, right hand and feet commands. Meanwhile, the EEG data with class labels are recorded. The experiments conducted in different days for the same subject are called different sessions.

Table 1: Timing scheme for paradigm



Data recording

In this data sets, the two devices of g.tec (g.USBamp) and Neuroscan (SynAmps RT) were used for recording the EEG signals. The EEG signals were band-pass filtered between 2Hz and 30Hz with sample rate of 256Hz and a notch filter at 50Hz was enabled for g.tec whereas the band-pass filter between 0.1Hz and 100Hz with sample rate of 250Hz was applied for Neuroscan device. The signals are measured in µV and V for Neuroscan and g.tec respectively.

The number of electrodes was different in the data sets, the configuration of 5, 6 and 14 channels were used in different data sets. This aims to develop the BCI system with electrode number as fewer as possible. The electrodes montage is shown in Figure 2. The green and blue electrodes were used in data set with 14 channels, which mainly focus on the motor and sensorimotor area, the blue electrodes were used in data set with 6 channels. For 5 channels data set, the electrodes of C3, Cp3, C4, Cp4, Cz were used to record the EEG signals.

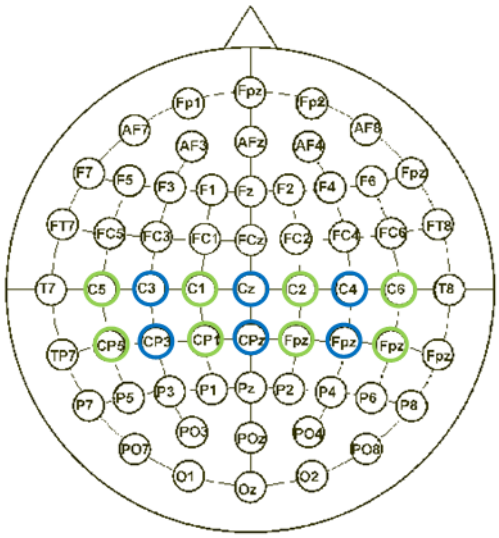


Figure 2: Electrodes positions

Data files and format

All data sets are stored in the Matlab format (*.mat). The file name consists of subject ID, channel number, imagery tasks and session number. For example,

'SubC_6chan_3LRF_s1': Subject C, 6 channels, 3-class imagery tasks of left hand, right hand and feet and session 1.

Each file contains one session which is comprised of several runs separated by short breaks. Some subjects has many sessions conducted in different days. It can be used to evaluate whether the classifier model trained from one session is able to classify the next session correctly.

The EEG data stored in each data set is organized in segment structure, each segment represents a single trial with one specific class label. The variables in each data set are:

- EEGDATA: The 3-way array with size of [*channel* x *time* x *trial*]. *'Channel'* denotes the number of electrodes, *'time'* is the duration of each imagination task and *'trial'* is the number of motor imagery tasks performed in this session.
- LABELS: vector of target classes (1,2,3) corresponding to each trial in variable 'EEGDATA'. The length of LABELS equals to the length of 3rd-mode of variable 'EEGDATA'.
- Info: structure providing additional information with fields
    - srate: sampling rate.
    - class: cell array of the names of the motor imagery tasks.
    - channels: cell array of channel labels.

Table 1 provides the information for each data set file including subject ID, motor imagery class , channel number, duration of each imagination task, trial number, sample rate, device name and the 10x10 folder cross validation performance (accuracy ± standard deviation) on this data set. Please note that this performance is roughly obtained by basic preprocess, CSP feature extraction and LDA classifier. The complete same method with same parameters are used to test all data sets without selecting the optimal channels, frequency band and feature number for each subject's data set. Therefore, this results can be used as a general comparison between these data sets and is helpful for understanding which data set is the best and which subject is the best.

Each file is recorded from an experiment which is independent from others. This means we can obtain more data sets with 2-class mental tasks for subject A, B and C by extracting the subset corresponding to the first two classes (left hand & right hand) from 3-class data set of the same subject. Hence, the performance of first two classes is also provided for the 3-class data sets. In table 1, the column of *class* denotes the different combination of mental tasks:

- LH/RH: 2 classes of left hand and right hand;
- LH/RF: 2 classes of left hand and right foot;
- LH/F: 2 classes of left hand and both feet;
- LH/RH/F: 3 classes of left hand, right hand and both feet;
- LH/RH/R: 3 classes of left hand, right hand and relaxation.

Table 1: The detail information of data set

2 classes								
Dataset	**Subject**	**Class**	**Channel**	**Duration (sec)**	**Trial number**	**10x10 CV (Acc.±std.)**	**Sample rate**	**Device**
SubA_6chan_2LR_s1	A	LH/RH	6	3s	130	0.88±0.01	256Hz	g.tec
SubA_6chan_2LR_s2					134	0.84±0.01		
SubB_6chan_2LR	B	LH/RH	6	4s	162	0.88±0.01	250Hz	Neuroscan
SubC_6chan_2LR_s1	C	LH/RH	6	3s	170	0.86±0.01	256Hz	g.tec
SubC_6chan_2LR_s2				3s	158	0.85±0.01		
SubC_6chan_2LR_s3				5s	48	0.92±0.01		
SubC_6chan_2LR_s4				3s	120	0.89±0.01		
SubC_6chan_2LR_s5				3s	90	0.93±0.01		
SubD_5chan_2LR	D	LH/RH	5	4s	80	0.75±0.02	256Hz	g.tec
SubE_5chan_2LR	E	LH/RH	5	4s	48	0.86±0.02	256Hz	g.tec
SubF_6chan_2LR	F	LH/RH	6	4s	80	0.71±0.03	256Hz	g.tec
SubG_6chan_2LR	G	LH/RH	6	4s	120	0.81±0.01	256Hz	g.tec
SubH_6chan_2LR	H	LH/RH	6	3s	150	0.71±0.02	256Hz	g.tec
SubA_6chan_2LF	A	LH/RF	6	3s	150	0.84±0.004	256Hz	g.tec
SubC_6chan_2LF_s1	C	LH/F	6	3s	330	0.96±0.002	256Hz	g.tec
SubC_6chan_2LF_s2					180	0.91±0.004		
SubC_6chan_2LF_s3					102	0.86±0.02		
3 classes								
Dataset	**Subject**	**Class**	**Channel**	**Duration (sec)**	**Trial number**	**10x10 CV (Acc.±std.)**	**Sample rate**	**Device**
SubA_5chan_3LRF	A	LH/RH/F	5	4s	270	0.92±0.004 0.91±0.03(2c)	256Hz	g.tec
SubB_5chan_3LRF	B	LH/RH/F	5	4s	174	0.86±0.01 0.92±0.01(2c)	250Hz	Neuroscan
SubB_6chan_3LRF			6		150	0.80±0.03 0.94±0.01(2c)		
SubC_5chan_3LRF	C	LH/RH/F	5	4s	180	0.86±0.01 0.90±0.04(2c)	256Hz	g.tec
SubC_6chan_3LRF_s1			6	3s	300	0.89±0.01 0.92±0.004(2c)		
SubC_6chan_3LRF_s2					300	0.84±0.01 0.87±0.01(2c)		
SubC_6chan_3LRF_s3					204	0.89±0.01 0.92±0.01(2c)		
SubC_5chan_3LRF _Day1	C	LH/RH/F	5	4s	210	0.72±0.02 0.78±0.02(2c)	256Hz	g.tec
SubC_5chan_3LRF _Day2					210	0.81±0.01 0.88±0.01(2c)		
SubC_5chan_3LRF _Day3					180	0.81±0.01 0.86±0.02(2c)		
SubC_5chan_3LRF _Day4					180	0.83±0.02 0.90±0.01(2c)		
SubC_5chan_3LRF _Day5					234	0.87±0.01 0.96±0.002(2c)		
SubC_5chan_3LRF _Day6					150	0.88±0.01 0.92±0.01(2c)		
SubC_5chan_3LRF _Day7					180	0.88±0.01 0.94±0.01(2c)		
SubC_14chan_3LRR	C	LH/RH/R	14	4s	350	0.78±0.01 0.88±0.004(2c)	250Hz	Neuroscan